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REVIEW

Regeneration or Risk? A Narrative Review of BPC-157 for Musculoskeletal Healing

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Abstract

Purpose of Review This scoping review aims to evaluate the molecular mechanisms, therapeutic potential, and safety concerns of Body Protective Compound-157 (BPC-157) in the context of musculoskeletal healing. Given the compound's increasing availability, popularity, and its regulatory controversies, we sought to assess the breadth and quality of preclinical and clinical data supporting its use in musculoskeletal medicine.

Recent Findings BPC-157 is a synthetic pentadecapeptide originally isolated from gastric juice and has demonstrated regenerative properties across numerous animal models. It activates several overlapping pathways, notably VEGFR2 and nitric oxide synthesis via the Akt-eNOS axis, promoting angiogenesis, fibroblast activity, and neuromuscular stabilization. It also engages ERK1/2 signaling, facilitates endothelial and muscle repair, and exerts anti-inflammatory effects. These effects promote angiogenesis, fibroblast activity, and neuromuscular stabilization, particularly in poorly vascularized tissues such as tendons and myotendinous junctions. Despite broad preclinical support, human data are extremely limited. Only three pilot studies have examined BPC-157 in humans, including its use for intraarticular knee pain, interstitial cystitis, and intravenous safety/pharmacokinetics. No adverse effects were reported, but rigorous, large-scale trials are lacking.

Summary BPC-157 demonstrates robust regenerative and cytoprotective effects in preclinical studies, positioning it as a potentially valuable tool in musculoskeletal medicine. Despite its growing popularity among athletes and its wide availability through non-regulated sources, there is minimal human data available. Until well-designed clinical trials are conducted, BPC-157 should be considered investigational, and its use approached with caution. This review highlights that given the robust preclinical evidence and high public interest, there is a critical need for well-designed human trials to assess the safety, efficacy, and clinical utility of BPC-157 in musculoskeletal medicine.

Keywords Pentadecapeptide · Scoping review · Musculoskeletal medicine · Research gaps

Introduction

Peptide therapy is an emerging area of research in musculoskeletal medicine. Peptides, naturally occurring short chains of amino acids synthesized in the body, act as selective signaling molecules that bind to specific cell surface receptors, triggering intracellular cascades [1]. Peptides play a crucial role in various biological functions and have been widely applied in drug development over the past decade [2]. Given their ability to regulate physiological mechanisms, research has increasingly focused on harnessing naturally-occurring peptides to develop new therapeutic tools [3]. Among peptides studied for tissue and muscle repair [4, 5], BPC-157 (also known as PLD-116, PL-10, PL14736, or Bepectin [6–9]) has gained attention [10, 11]. Originally isolated from human gastric juice by Dr. Predrag Sikiric in 1993,

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BPC-157 has demonstrated regenerative properties in animal models, promoting the healing of tendons, ligaments, muscles, nerves, bones, teeth, corneas, and the endothelium [15–29]. Studies also suggest it may be hepatoprotective, aid recovery from traumatic brain injury, and influence blood clot formation/degradation [27–33].

Due to its potential musculoskeletal benefits, including injury prevention, enhanced exercise recovery, and tissue repair [5, 20, 24], BPC-157 has gained anecdotal popularity among athletes, with increasing reports of its use in sports and fitness communities over the past 10–15 years through medical clinics, compounding pharmacies, and direct-to-consumer sales [34]. However, in 2022, WADA banned BPC-157 under the S0 Unapproved Substances category, prohibiting its use in both competition and training due to its lack of human clinical approval [35, 36]. In September 2023, the FDA further restricted its use by classifying it as a Category 2 bulk drug, effectively barring its inclusion in compounded medications due to concerns about safety, impurities, and insufficient human data [37]. While this does not constitute an outright ban, it effectively prohibits its legal use in compounded medications under current FDA regulations. Despite these regulatory actions, BPC-157 remains widely available for human use through various sources. This review will explore the molecular pathways of BPC-157, its potential effects on performance, its role in injury healing, its applications beyond musculoskeletal conditions, and the literature on both reported and theoretical side effects.

Methods

A structured literature search was conducted by the research team with the assistance of an academic librarian, with searches of six electronic databases (PubMed, Embase [Elsevier], CINAHL Complete [Ebscohost], SportDiscus [Ebscohost], BIOSIS Citation Index, and Web of Science Core Collection [Clarivate]) for studies related to BPC-157 published through March 2025. Search strategies incorporated combinations of keywords and subject headings including "BPC-157," "PLD-116," "PL-10," "PL 14736," and "body protective compound," and were tailored to each database. Filters were applied to limit results to English-language, peer-reviewed publications. A comprehensive search strategy was also conducted in PubMed using both MeSH terms and text words to capture studies involving human subjects and review articles.

Three reviewers independently screened titles and abstracts for relevance to musculoskeletal applications, molecular mechanisms, performance outcomes, and side effect profiles. Full texts were retrieved for studies meeting inclusion criteria. Discrepancies in study selection were resolved by consensus. Reference lists of key articles were manually reviewed to identify additional relevant sources.

Molecular Pathways of BPC-157 in Musculoskeletal and Neuromuscular Systems

BPC-157 exerts regenerative effects through multiple interconnected pathways, facilitating musculoskeletal and neuromuscular healing (Fig. 1). Its ability to modulate angiogenesis, inflammation, nitric oxide signaling, neurotransmitters, and tissue-specific repair highlights its pleiotropic nature [20, 23–26, 30, 38–41].

Molecular Pathways

BPC-157 significantly promotes angiogenesis by enhancing vascular endothelial growth factor receptor-2 (VEGFR2) activity and nitric oxide (NO) signaling primarily through activation of the Akt-endothelial nitric oxide synthase (eNOS) pathway. This pathway increases NO production, essential for endothelial proliferation, vessel dilation, and new capillary formation, which may be particularly beneficial in ischemic or hypovascular musculoskeletal tissues [20, 24, 26, 39, 41]. Furthermore, BPC-157 stabilizes existing vascular structures and modulates vascular tone via NO-mediated vasodilation, thereby safeguarding tissues against ischemic damage during repair processes [20, 24, 41]. Additionally, BPC-157 exerts cytoprotective effects by enhancing eNOS activity through Src kinase-caveolin-1 signaling. This mechanism supports cellular resilience by upregulating endogenous antioxidants, including heme oxygenase-1 (HO-1), thus reducing oxidative stress, preventing mitochondrial dysfunction, and limiting apoptosis [11, 20, 23, 41, 42].

BPC-157 also preserves neuromuscular function by stabilizing acetylcholine receptors and nerve terminals at the neuromuscular junction (NMJ), effectively reversing paralysis induced by neuromuscular blockers such as succinylcholine and lidocaine [24, 43, 44]. It further maintains central nervous system homeostasis by normalizing disrupted neurotransmitter signaling involving dopamine, serotonin, and gamma-aminobutyric acid (GABA), essential for stabilizing synaptic function and ensuring neuromuscular coordination after injury or toxin exposure [43, 45]. BPC-157 has also demonstrated the ability to normalize glutamatergic signaling, including in models of NMDA receptor overactivation [41]. By counteracting the neurotoxic effects of agents like ketamine and MK-801, BPC-157 may restore excitatory neurotransmission and synaptic plasticity following injury or pharmacologic disruption [41, 46, 47].

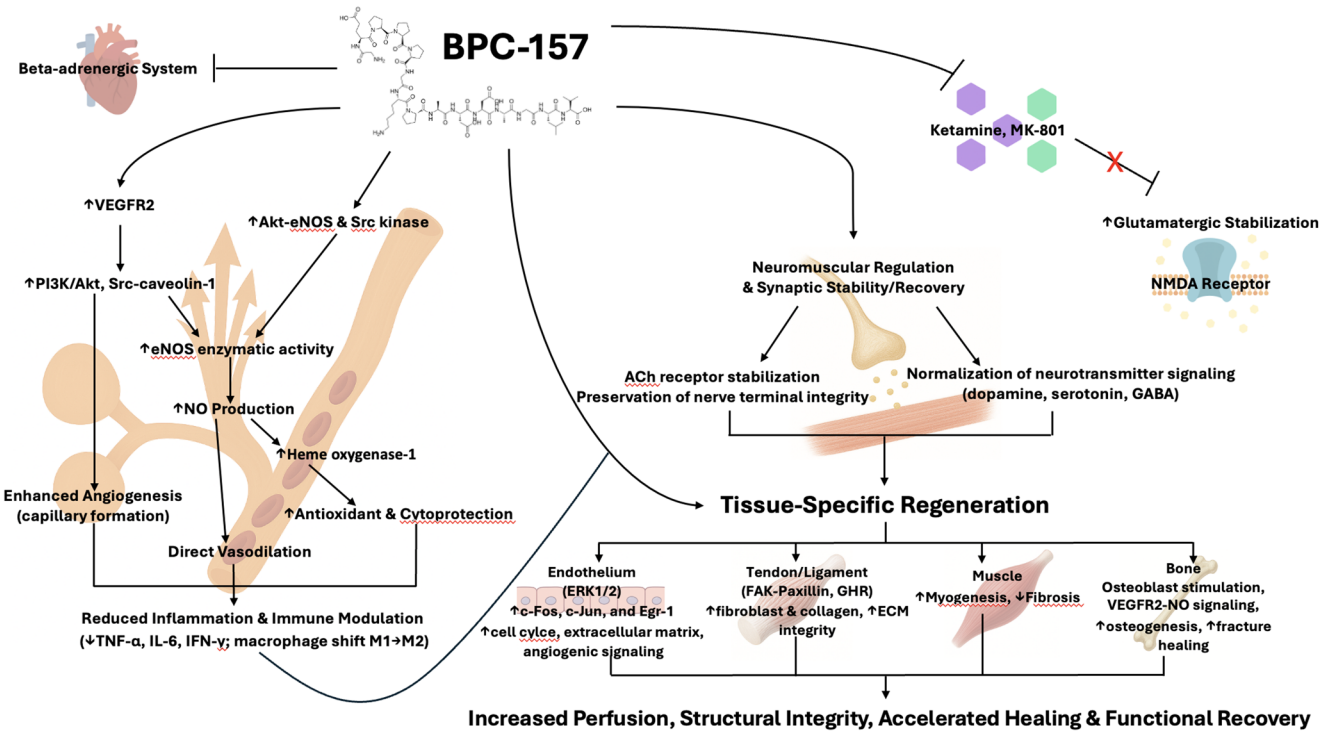


Fig. 1 Diagram illustrating the molecular mechanisms by which BPC-157 promotes tissue regeneration, cytoprotection, and neuromodulation. BPC-157 activates VEGF-dependent (via VEGFR2–PI3K–Akt–eNOS) and VEGF-independent (via Src–caveolin-1–eNOS) pathways to NO production, supporting angiogenesis, vasodilation, and vascular stability. It upregulates cytoprotective factors such as heme oxygenase-1 (HO-1) and heat shock proteins, preserving mitochondrial integrity and reducing oxidative stress. BPC-157 also restores glutamatergic signaling after NMDA receptor overactivation (e.g., ketamine, MK-801 exposure) and modulates adrenergic balance by counteracting beta-adrenergic overstimulation and blockade, preventing vascular occlusion-like syndromes. In endothelial cells, BPC-157 activates ERK1/2 signaling, enhancing proliferation, migration, and vascular tube formation through transcription factors like c-Fos, c-Jun, and Egr-1; ERK1/2 activation is required for its pro-healing effects both in vitro and in vivo, including in alkali-burn wound models

Additionally, BPC-157 modulates the adrenaline and noradrenaline systems by counteracting both beta-adrenergic overstimulation (e.g., isoprenaline) and blockade (e.g., sotalol), preventing resultant hemodynamic collapse, multiorgan failure, and vascular occlusion-like syndromes in rodent models [41, 48, 49].

Finally, BPC-157 demonstrates pronounced anti-inflammatory properties by significantly decreasing pro-inflammatory cytokines tumor necrosis factor-alpha (TNF-α), interleukin-6 (IL-6), and interferon-gamma (IFN-γ). It promotes resolution of inflammation by shifting macrophage activity from a pro-inflammatory (M1) phenotype toward a reparative (M2) phenotype, thereby effectively reducing fibrosis and promoting enhanced tissue regeneration [11, 23, 26, 50].

While BPC-157 is noted to have a very short half-life (less than 30 min [51, 52]), the angiogenic, anti-inflammatory, and tissue regeneration processes it initiates appear to persist for weeks to months in animal studies, far exceeding the peptide's pharmacokinetic presence [19, 53]. Animal data demonstrates that BPC-157's therapeutic effects continue long after administration stops: for example, in spinal cord injury studies, functional improvements and reduced spasticity were maintained for up to 360 days after a single treatment [19], while in one tendon healing study, biomechanical improvements persisted through the 21–72 day observation period [53]. The sustained effects are thought to result from BPC-157's activation of multiple gene expression pathways within minutes of administration (including genes like Akt1, VEGFR2, eNOS, and numerous growth factors), which then trigger cascading cellular processes that continue independently. Essentially, the peptide acts as a biological "switch" that initiates self-sustaining healing programs rather than requiring continuous presence [23, 26, 39, 54].

Tissue-Specific Effects

Tendon and Ligament Healing

BPC-157 accelerates tendon and ligament repair through enhanced fibroblast proliferation and collagen synthesis, primarily via focal adhesion kinase (FAK)-paxillin signaling pathways [24, 26, 53]. It also increases growth hormone

receptor (GHR) expression in fibroblasts, augmenting the anabolic healing response, thus improving tendon structure and biomechanical function, even in conditions impaired by corticosteroids or limited vascular supply [15, 53].

Muscle Regeneration

In rodent models, BPC-157 significantly enhances myogenesis, muscle fiber regeneration, and functional recovery post-injury in skeletal muscle [38, 53]. By facilitating rapid re-establishment of myotendinous junctions and reducing fibrosis at injury sites, BPC-157 improves overall muscular integrity and contractile function, crucial for recovery from severe muscular trauma or degenerative muscular conditions [20, 26, 38].

Bone Healing

BPC-157 promotes osteogenesis and accelerates bone healing, particularly under compromised conditions such as delayed union, avascular osteonecrosis, or impaired fracture healing [20, 26, 53]. By stimulating angiogenesis within bone tissue and enhancing osteoblast activity via VEGFR2-NO signaling, BPC-157 markedly improves bone matrix deposition and fracture consolidation, highlighting its clinical promise for orthopedic conditions [20, 26].

Endothelium

BPC-157 activates the ERK1/2 (extracellular signal-regulated kinase) pathway as a key mechanism underlying its pro-healing and angiogenic properties. In endothelial cell models, BPC-157 significantly enhances ERK1/2 phosphorylation in a dose-dependent manner, leading to increased cellular proliferation, migration, and vascular tube formation—processes central to tissue regeneration and angiogenesis [26, 55]. These effects are mediated through downstream activation of transcription factors such as c-Fos, c-Jun, and EGR-1, which regulate genes involved in cell cycle progression, extracellular matrix remodeling, and angiogenic signaling [26]. In both in vitro and in vivo rodent studies, BPC-157 induces early expression of EGR-1 and its corepressor NAB2, forming a regulatory feedback loop that modulates the duration and amplitude of angiogenic gene transcription during wound healing, which suggests BPC-157 may engage feedback mechanisms to prevent excessive angiogenic signaling [56]. The functional significance of ERK1/2 activation by BPC-157 is further supported by inhibitor studies: when the ERK pathway is pharmacologically blocked, BPC-157's pro-migratory and pro-angiogenic effects are abolished, underscoring the pathway's critical role in mediating cellular activity [55]. In rodents, BPC-157 accelerates healing in alkali-burn wound models, coinciding with increased ERK1/2 activity in regenerating tissues, suggesting that this signaling axis contributes not only to cellular dynamics in vitro but also to tissue-level repair responses in vivo [26, 55].

Musculoskeletal Healing

Injury healing typically progresses through three phases: the inflammatory phase (days 1–5), the repair/proliferative phase (days 5–14), and the remodeling phase (days 14–90+) [57]. During the inflammatory phase, inflammatory cells infiltrate the injury site, followed by fibroblast proliferation and the synthesis of collagen fibers in the repair phase. The remodeling phase sees the maturation and alignment of these fibers to restore tissue integrity. BPC-157 has shown potential in accelerating the healing process by targeting these key stages, particularly in tissues with limited blood supply (i.e. tendons and ligaments) [24, 58]. Cerovecki et al. demonstrated healing of the medial collateral ligament in rats after surgical transection [15]. Perovic et al. suggested this peptide has the ability to impact healing of spinal cord injury by targeting all stages of the secondary injury phase [19]. Through its angiogenic properties, BPC-157 promotes the growth of new blood vessels, significantly enhancing the healing of the myotendinous junction, which is typically a weak spot due to its poor vascularization [24, 38]. In rats with hind limb ischemia, BPC-157 both increased the vessel density and accelerated blood flow recovery [39]. Studies have demonstrated that BPC-157 not only improves tendon healing but also enhances tendon-to-bone integration, even in the presence of corticosteroids, which is notable due to steroids usual effect impairing recovery [53, 59]. This effect extends to muscle injuries, where BPC-157 has shown to restore muscle function and integrity, even after traumatic injuries or corticosteroid application [16]. The majority of musculoskeletal animal studies dose BPC-157 once daily, with regimens ranging from a single dose to daily administration for 7, 14, 28, or up to 90 days, depending on the injury model and outcome measures [20, 24, 26, 38, 60, 61]. However, while animal studies have shown promising results, it is important to recognize that there is still a lack of extensive human studies to fully assess the impact of BPC-157 on musculoskeletal injuries in humans [51, 62].

Performance

BPC-157 may also indirectly influence athletic performance by improving recovery processes. While there have been no direct human studies measuring its effects on speed or strength in athletes, BPC-157 could potentially allow athletes to adapt to training stimuli more effectively, given its

effects related to accelerated healing and tissue repair [11, 63, 64]. In one study comparing control and sham rats, BPC-157-treated rats demonstrated significantly improved recovery, as evidenced both biochemically and through micro- and macroscopic observations [11]. Biomechanically, the healed tendons showed an increased load to tendon failure and significantly higher functionality, as indicated by the Achilles Functional Index (AFI) over 14 days. This suggests that BPC-157 may improve tissue integrity and function, which could, in turn, enhance an athlete's ability to recover and perform. Given that post-exercise recovery is critical for preventing metabolic disturbances and injury, which can impair performance, the peptide's potential in this area is promising. While BPC-157 may help optimize cellular pathways affected by exercise, the extent of its impact on performance and whether its use is both safe and ethical for athletes remains in question.

Human Trials

Unfortunately, there have been limited in vivo clinical trials conducted to demonstrate the effectiveness of BPC-157 usage in humans. To our knowledge, there have only been three published studies. In a 2021 retrospective study, Lee and Padgett compared the effectiveness of knee injections using BPC- 157 alone to knee injections using a combination of BPC-157 and thymosin-beta-4 (TB4), which is another peptide shown to have anti-inflammatory properties. Sixteen patients were contacted to review their knee pain symptom improvements 6 months to 1 year after receiving the injections. In total, 14 of the 16 patients had significant pain relief when given an intraarticular knee injection with BPC-157 or BPC-157 plus TB4. While this study had many limitations (e.g. small sample size, lack of control group, lack of common diagnosis), it did demonstrate improved pain in 87.5% of patients who received injections containing BPC-157 [34]. In another pilot study of 12 individuals, conducted by Lee et al. (2024), intravesicular (bladder) injections of BPC-157 were reported to result in 80–100% resolution of moderate to severe interstitial cystitis at 6 weeks post- treatment [65]. Notably, all patients had previously been treated with pentosan polysulfate, which is currently the only FDA approved treatment for interstitial cystitis, and had not responded. The Global Response Assessment Questionnaire was utilized to evaluate the patients' perception of the effectiveness of this therapy and all 12 patients responded with a score corresponding to "significant improvement". Additionally, cystoscopic imaging done pre- and post-treatment demonstrated resolution of detrusor hyperemia, hypertrophy, and hypervascularity in at least one patient. Although the exact mechanism of interstitial cystitis is unknown, there is thought to be a significant contribution from chronic inflammation, therefore BPC-157 was chosen in this study due to its anti-inflammatory properties and its role in restoring damaged epithelium as demonstrated in various animal models [22, 66, 67]. Most recently, Lee and Burgess (2025) conducted a pilot study involving two healthy adults who received intravenous BPC-157 infusions up to 20 mg. The treatment was well tolerated, with no adverse events or clinically meaningful changes observed in vital signs, electrocardiograms, or laboratory biomarkers assessing cardiac, hepatic, renal, thyroid, or metabolic function. Pharmacokinetic analysis showed that plasma BPC-157 concentrations returned to baseline within 24 h, consistent with its known rapid clearance and short half-life [52].

Side Effects

Due to the paucity of studies done in humans thus far, there is also limited data regarding safety and associated side effects. In the three human studies completed, the safety profile has been promising. In the cystitis study by Lee et al. (2024) discussed above, there were no reported side effects after intravesicular BPC-157 administration. Specifically, the participants were screened for fevers, skin rash, nausea, vomiting, worsening urinary symptoms and dyspareunia. Additionally, 0 of 12 participants experienced any hematuria or acute cystitis following treatment [65]. The study conducted by Lee and Padgett (2021) with intraarticular injections of BPC-157 did not report any adverse effects, although the screening for adverse events was not discussed at length [34]. The 2025 Lee and Burgess pilot study in two healthy adults showing that intravenous BPC-157 was well tolerated with no adverse effects and plasma levels returned to baseline within 24 h [52].

Despite the favorable tolerance observed in humans, proposed side effects have been discussed based on the in vitro effects of BPC-157. These possible unwanted effects include pathologic angiogenesis, toxic metabolite formation, and overproduction of nitric oxide [23]. Pathologic angiogenesis is implicated in the proliferation of tumor cells, as well as in immune and inflammatory disease processes [68, 69]. Angiogenesis occurs with contribution from VEGF and its receptors, eNOS and EGR-1, all of which have been shown to be affected and upregulated by BPC-157 [8, 38, 39, 70–72]. However, contrary to the tumor-promoting potential of many angiogenic agents, BPC-157 has been shown to inhibit uncontrolled cell proliferation, downregulate VEGF expression, and counteract VEGF-driven tumorigenesis, including suppression of Ki-67 and VEGF pathway activation [31, 73]. One of the primary metabolites of BPC-157 is thought to be Proline, which has been shown to play a role in superoxide production and intracellular cascades whose downstream effects could interfere with a variety

of biomolecules leading to widespread pathologic effects [23, 51, 74, 75]. Lastly, as previously discussed, BPC-157 robustly stimulates the NO system, which, at high levels, inhibits heme insertion into hemoglobin and alters the activity of heme thiolate enzymes and CYP enzymes [54, 76, 77]. These effects are associated with the development of anemia and altered drug metabolism [76]. Additionally, upregulation of NO has also been implicated in the development of neurodegenerative diseases such as Alzheimer's and Parkinson's disease [78–80]. Nonetheless, BPC-157 appears to selectively modulate angiogenesis depending on tissue context. For instance, it promotes corneal wound healing without neovascularization [22] and attenuates tumor cachexia by reducing IL-6, TNF-α, and cachexia-related pathways (FoxO3a, p-AKT, p-mTOR, P-GSK-3β) [50]. It also demonstrates protective antioxidant effects by lowering malondialdehyde (MDA), a cytotoxic lipid peroxidation product linked to carcinogenesis [81, 82].

Although the proposed adverse effects discussed above deserve scrutiny, a pre-clinical safety evaluation by Xu et al. (2020) found BPC-157 to be well tolerated at high doses in animal models. The researchers were unable to identify a minimum toxic dose or a lethal dose and reported no teratogenic, genotoxic, anaphylactic, or local toxic effects. These findings support the rationale for initiating formal human clinical trials [83].

Conclusion

BPC-157 demonstrates therapeutic potential across a broad range of musculoskeletal injuries through its angiogenic, anti-inflammatory, cytoprotective, and tissue-regenerative properties. Preclinical studies consistently highlight its ability to accelerate tendon, muscle, and ligament, healing, even under compromised conditions [15, 20, 24, 26, 39, 53]. Its proposed mechanisms, ranging from modulation of nitric oxide signaling to stabilization of neuromuscular junctions, suggest BPC-157 may significantly enhance recovery and functional outcomes in musculoskeletal injuries [20, 23, 25, 26, 43]. In addition to its regenerative effects, animal studies and one human pilot sutdy have shown BPC-157 to be well tolerated even at high doses, with no identified toxic or lethal thresholds and no observed teratogenic, genotoxic, or anaphylactic effects [52, 83]. This favorable preclinical safety profile has led investigators to advocate for the initiation of formal human trials [52, 83].

However, despite this promise, and growing interest in athletic and online communities, human data is exceedingly sparse [34, 52, 65]. The absence of high-quality trials in humans limits the ability to assess its human safety, efficacy, and appropriate clinical use. Furthermore, all the published studies report positive or beneficial effects of BPC-157, suggesting a possible publication bias toward positive findings, thus raising questions about the robustness and reproducibility of the reported benefits. This lack of human evidence has led both the FDA and WADA to restrict BPC-157, citing concerns about unregulated use, unknown human toxicity profiles, and potential for misuse [35–37]. Its continued availability via "research chemical" websites and the growing presence of anecdotal promotion on social media only heighten the urgency for rigorous scientific investigation. Until well-designed human trials are conducted and published, BPC-157 should not be recommended for clinical use in musculoskeletal medicine.

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This pilot study evaluated BPC-157 in patients with interstitial cystitis, making it one of the few human trials.

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The only available pharmacokinetic and safety study in healthy human volunteers using intravenous BPC-157.

Author Contributions All authors contributed to the conception and design of the review. F.M., A.L., and R.M. conducted the literature search and data synthesis. F.M. drafted the Introduction, Molecular Pathways, and Conclusion sections, prepared Figure 1, and integrated all sections of the manuscript. A.L. drafted the sections on Performance Literature and Injury Healing Literature. R.M. drafted the sections on Non-Musculoskeletal Conditions and Side Effects Literature. L.S. provided clinical expertise and contributed to final manuscript editing. D.C. supervised the project and critically reviewed all sections. All authors have read and approved the final manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Conflict of interest Flynn McGuire, Riley Martinez, Annika Lenz, Lee Skinner, and Dan Cushman declare that they have no conflict of interest.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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